HEAVY HAUL SEMINAR • APRIL 23-24, 2026

Current North American Railroad Wheel Service Issues – The Rise of Broken Flanges and the Importance of Wheel Handling to Prevent Plate Failures

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Presentation Outline

- Historical and current AAR MD115/CRB wheel failure data
- Wheel Failure Types

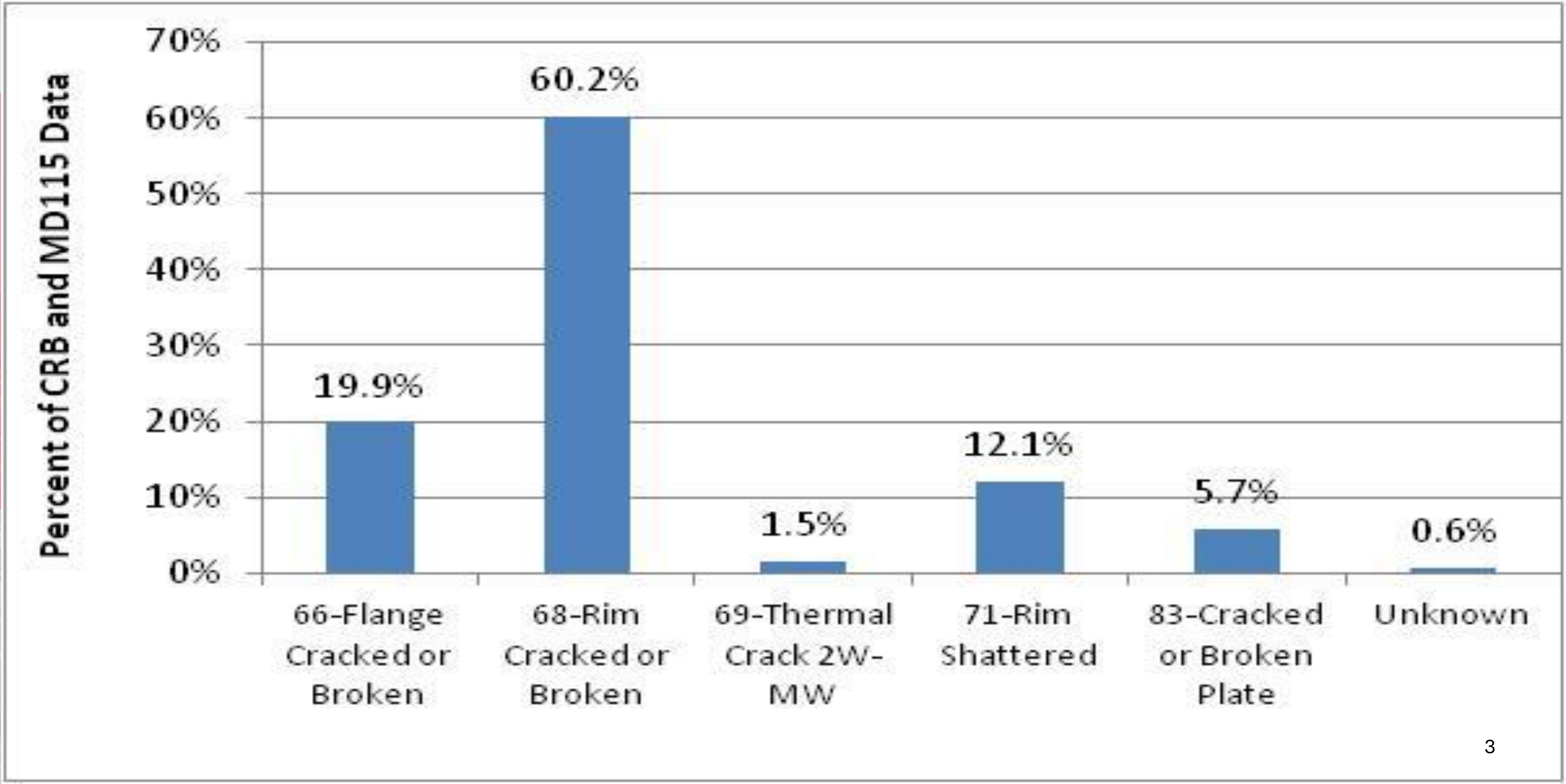
  WM 71 Shattered Rim

  WM 68 Cracked or Broken Rim (VSR 1D in particular)

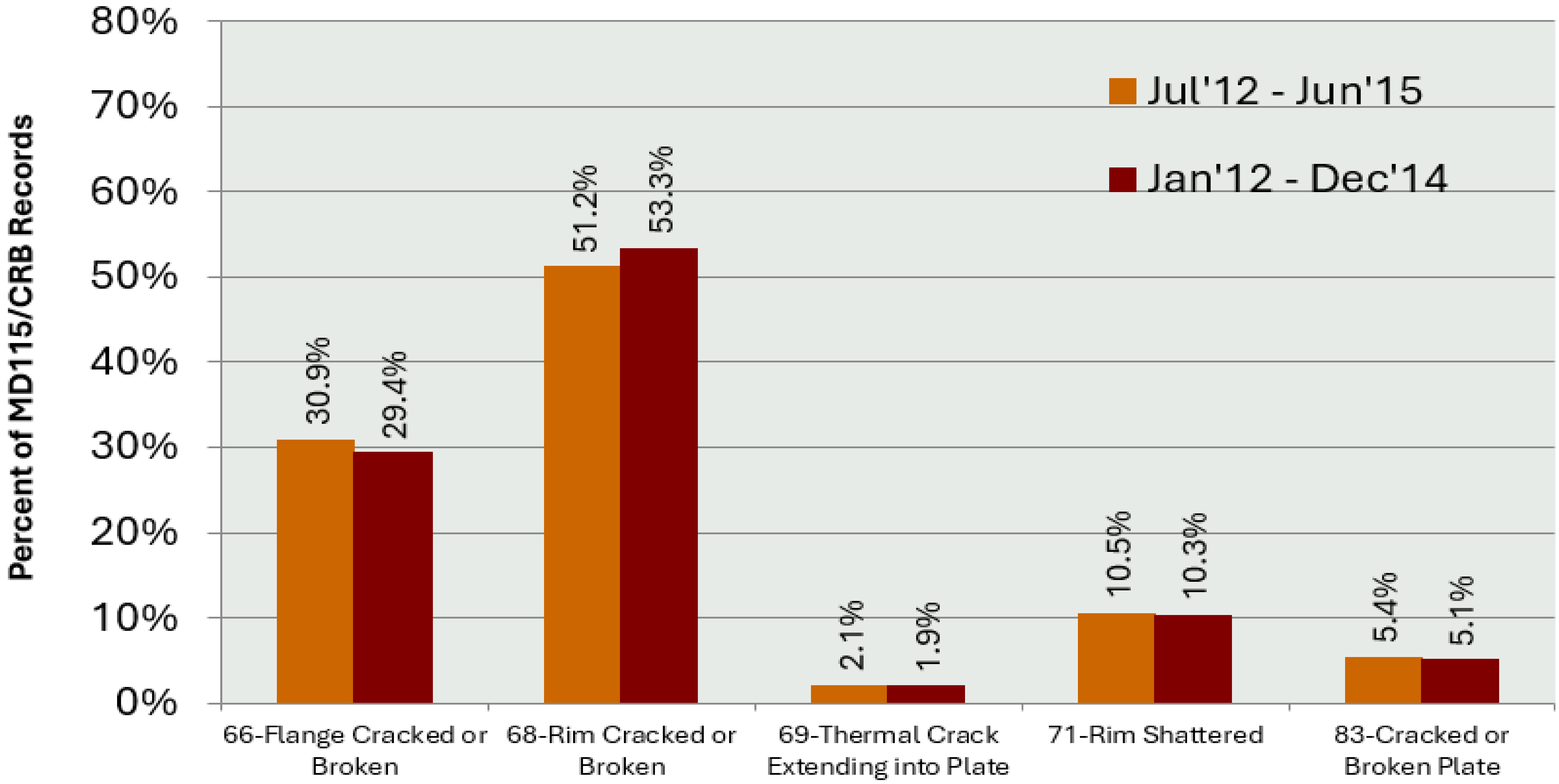
  WM 66 Flange Failures (Flange Face Shells, Flange Tip Origin)

  WM 83 Cracked or Broken Plate
- Wheel Plate Failures due to Impact Damage
- Wheel Plate Failures due to Weld Damage

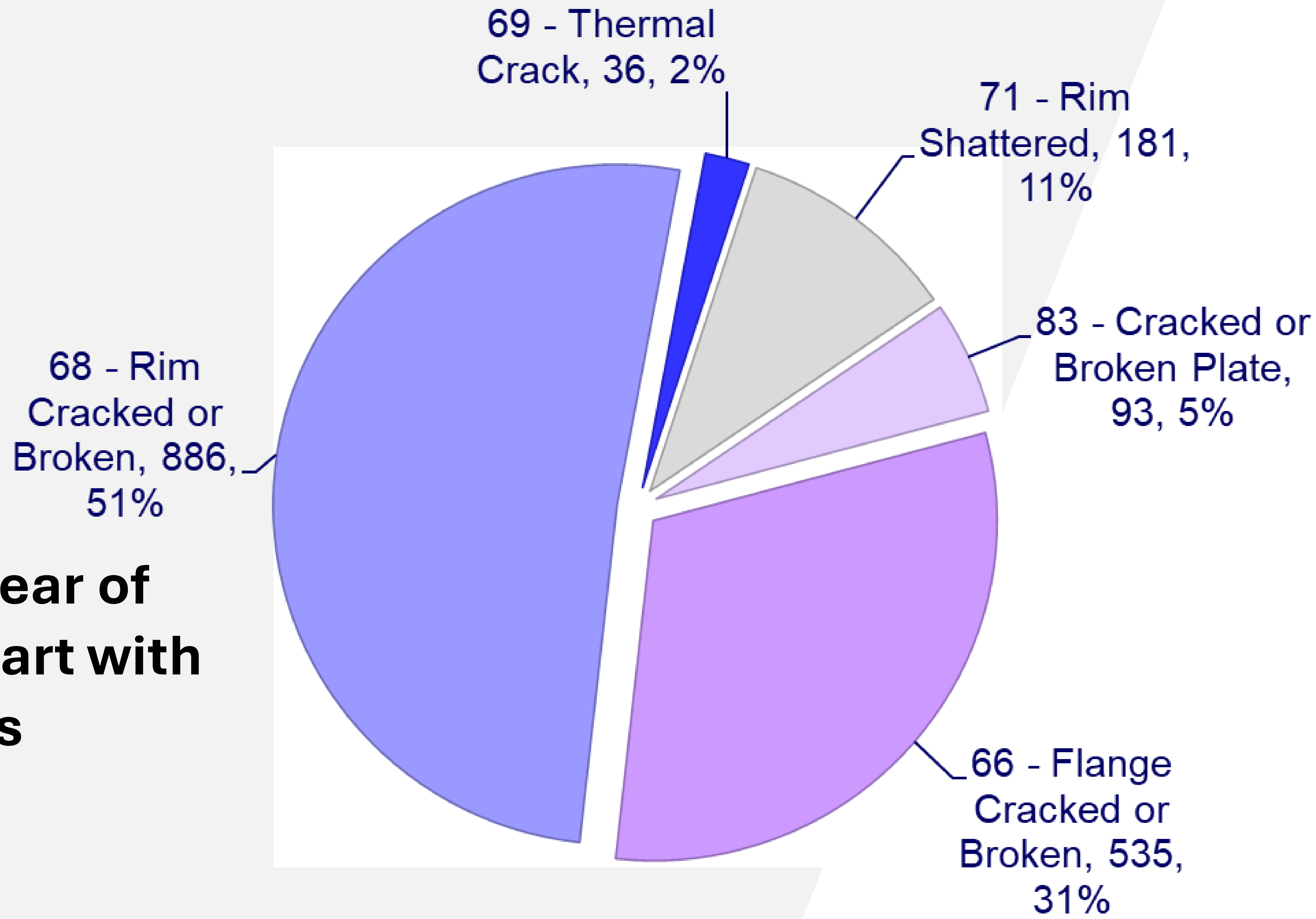
AAR MD115/CRB Data - August 2012



AAR MD115/CRB Data - August 2015

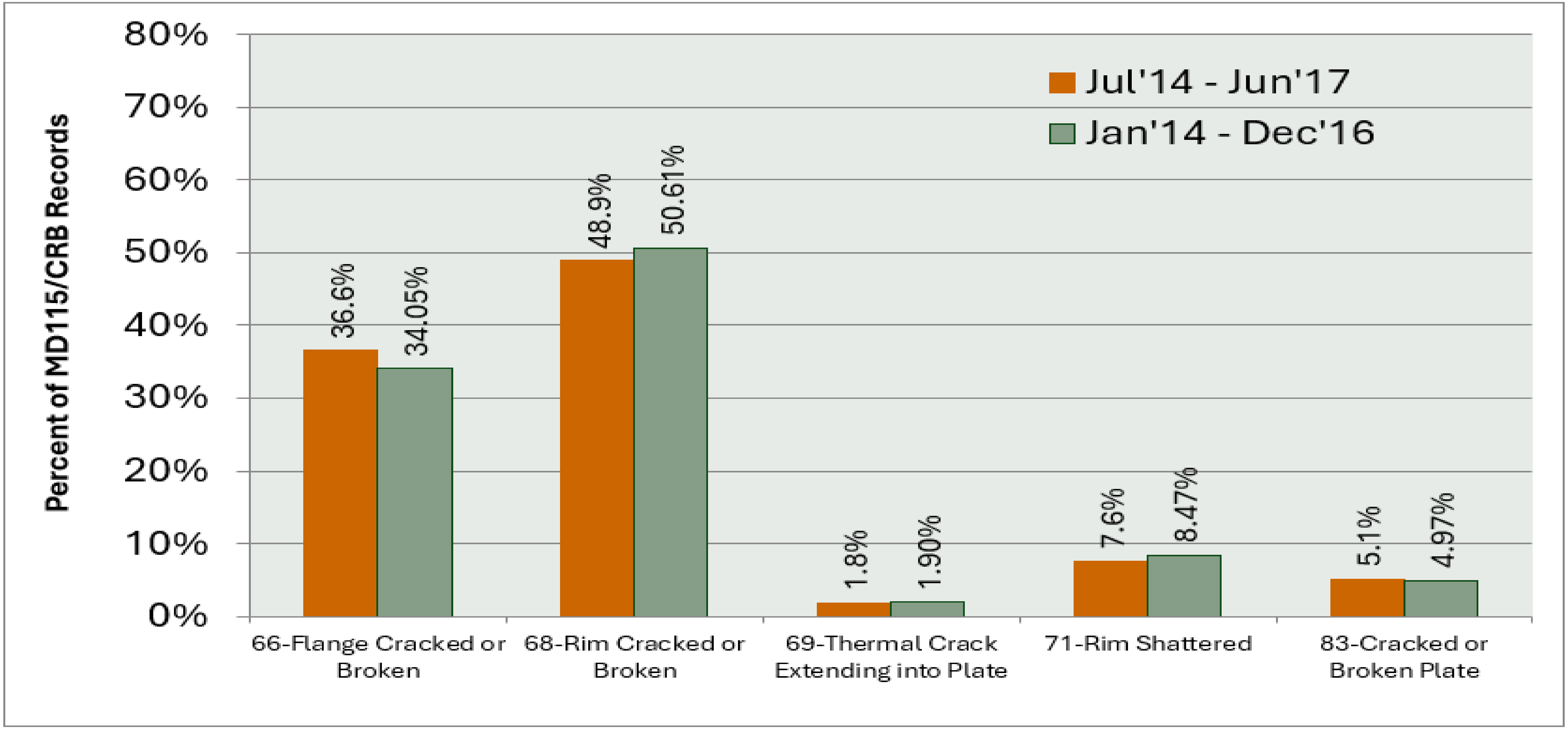


AAR MD115/CRB Data - August 2015

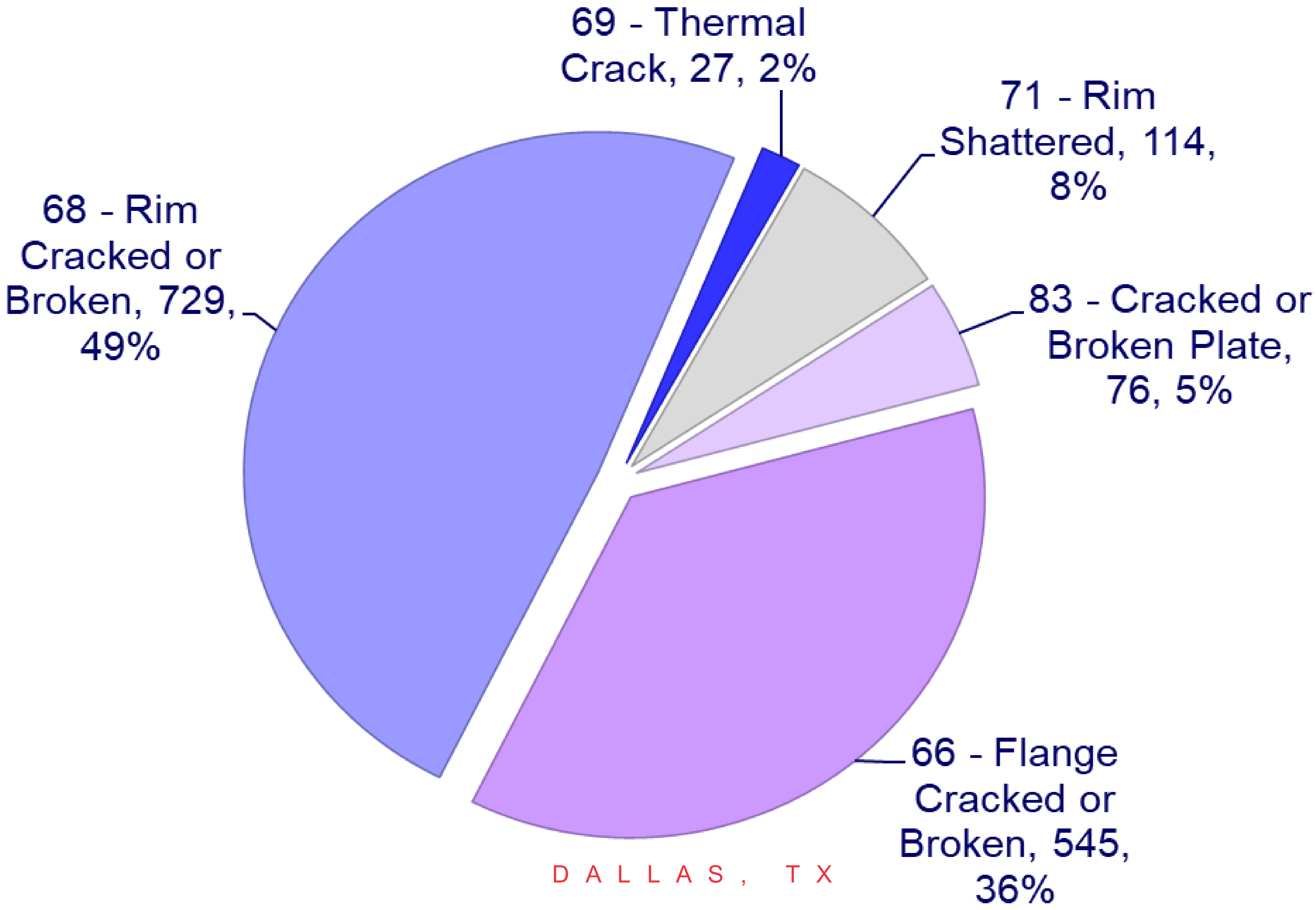


First year of pie chart with counts

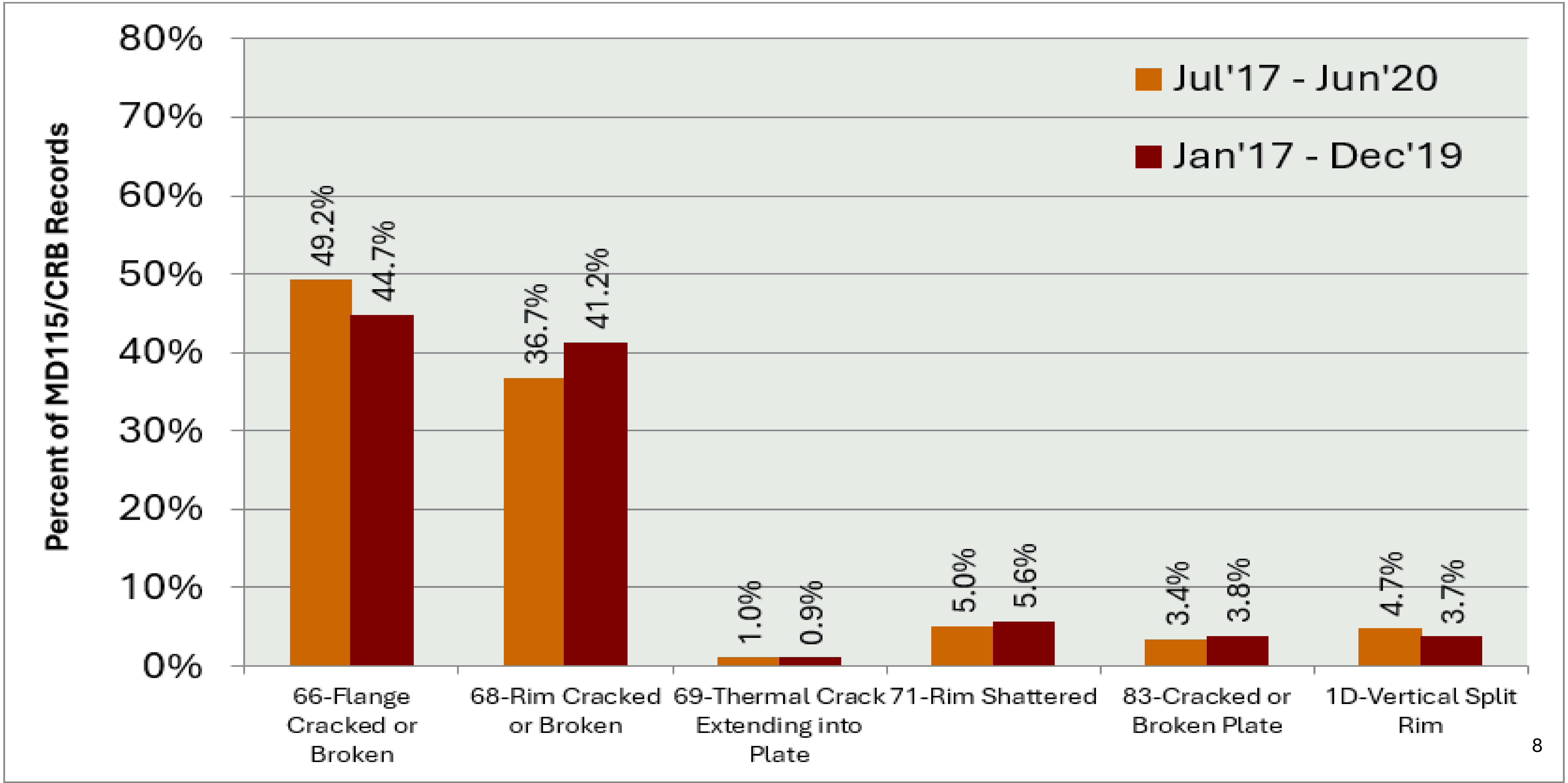
AAR MD115/CRB Data - August 2017



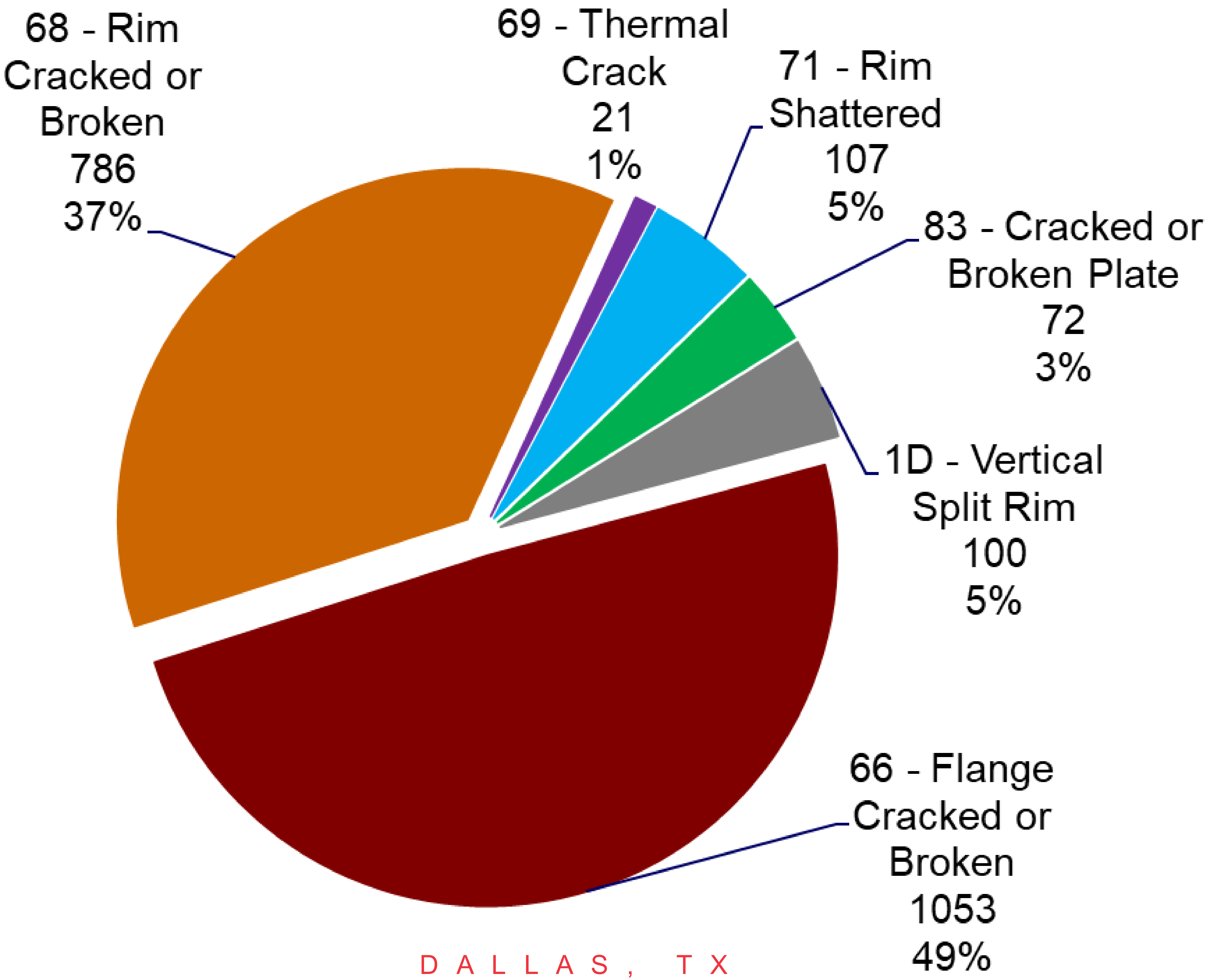
AAR MD115/CRB Data - August 2017



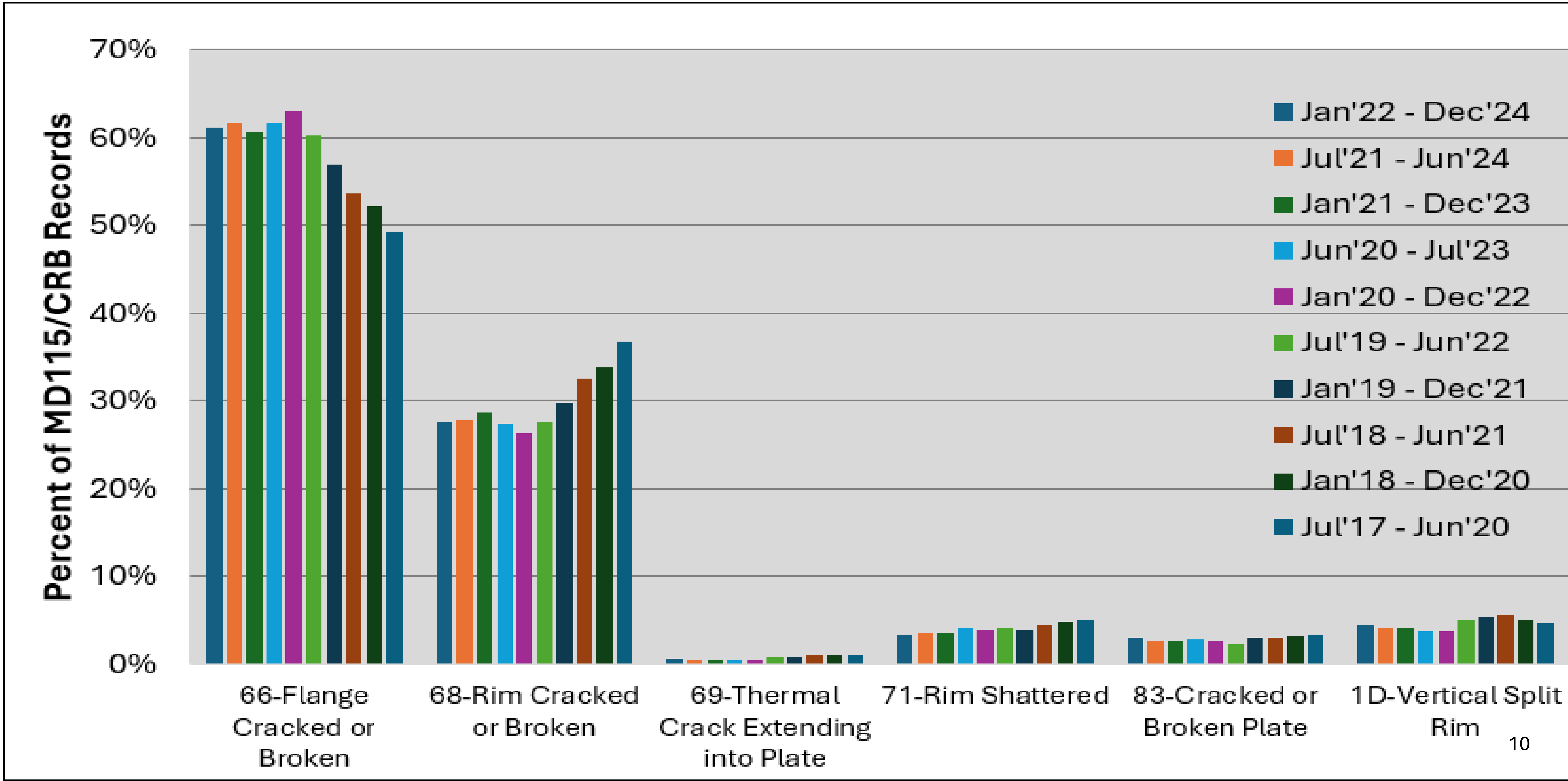
AAR MD115/CRB Data - August 2020



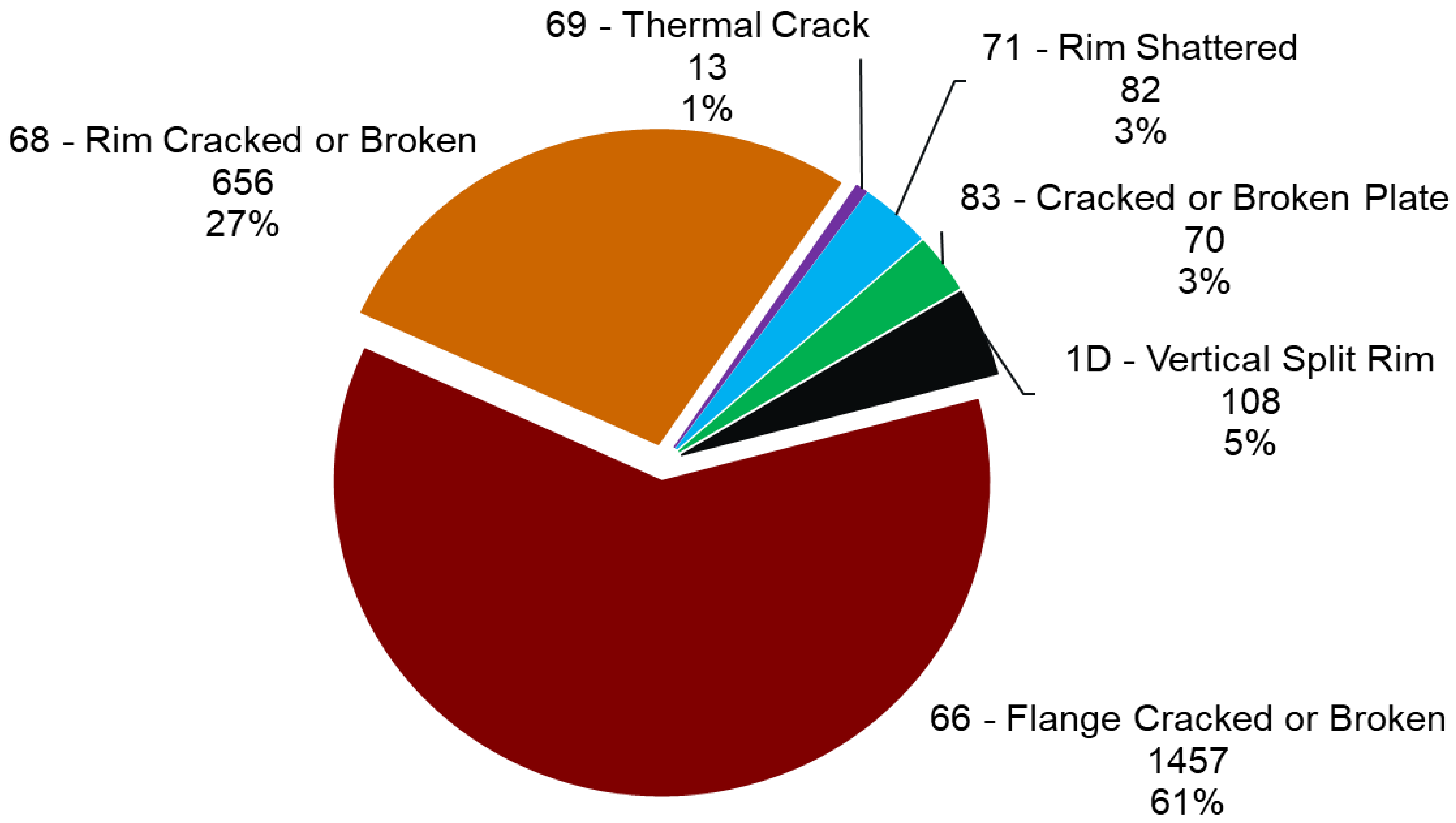
AAR MD115/CRB Data - August 2020



AAR MD115/CRB Data – January 2025



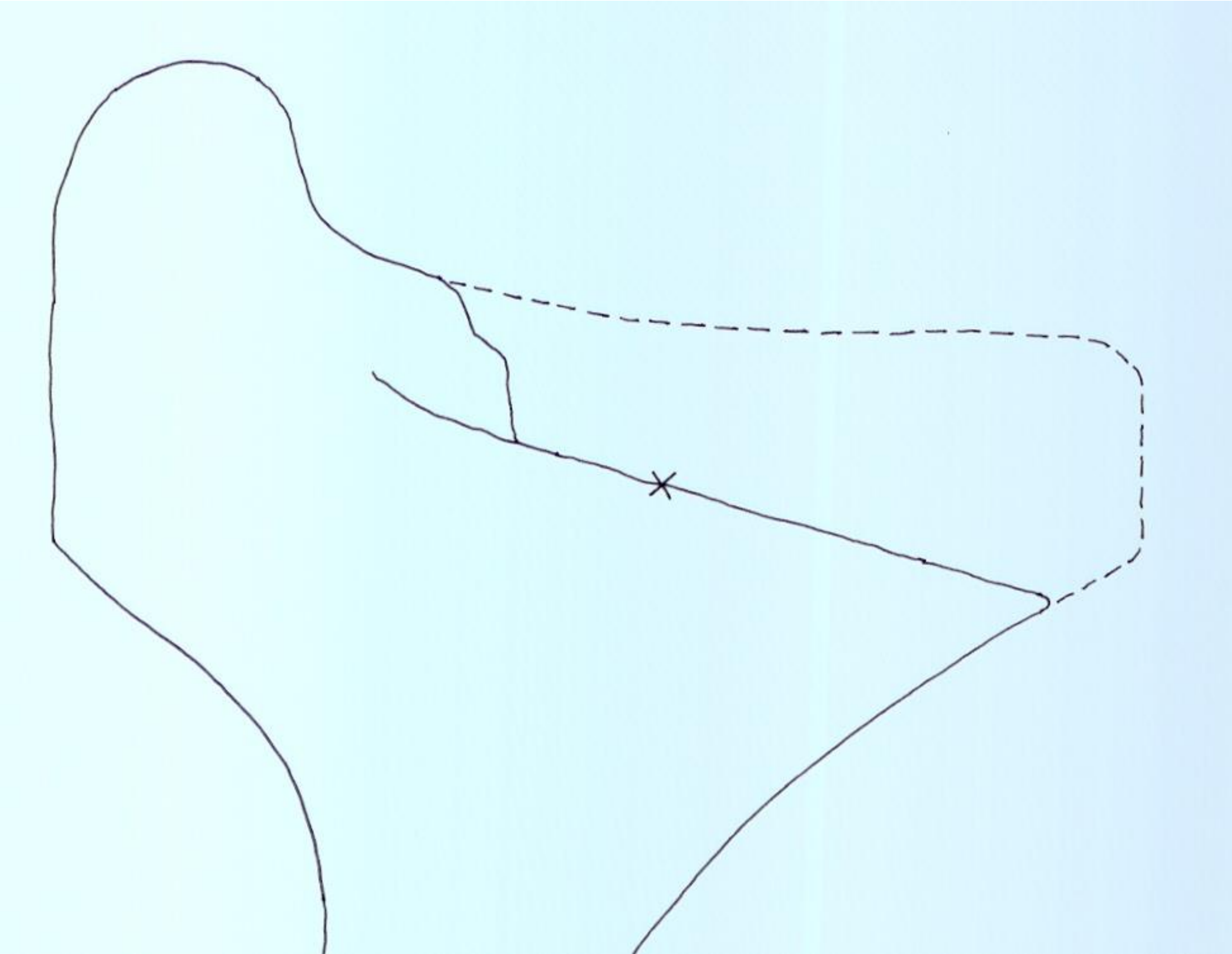
AAR MD115/CRB Data – January 2025



What the data tells us....

- Cracked/Broken Flanges have increased from 20% of broken wheels in 2012, to now 61%
  - (535 flanges in 2015 vs. 1,457 Jan 2025)
- Cracked or Broken Rims were 60% in 2012 to now 32% today - included new WM code 1D for Vertical Split Rims
  - (886 rims in 2015 vs. 764 Jan 2025)
- Cracked/Broken Plates have decreased from 5.7% to 3% Jan 2025
- Shattered Rims have decreased from 12% in 2012 to 3% Jan 2025
- Thermal crack failures remain rare

Shattered Rims WM 71 – Caused by subsurface porosity in cast wheels, and inclusions in forged wheels. Classic fatigue failures. This WM Code has decreased due to tighter AAR UT testing, adding a riser to cast 36” wheels (mid-1990’s), argon shrouding, and clean steel efforts by wheel manufacturers.

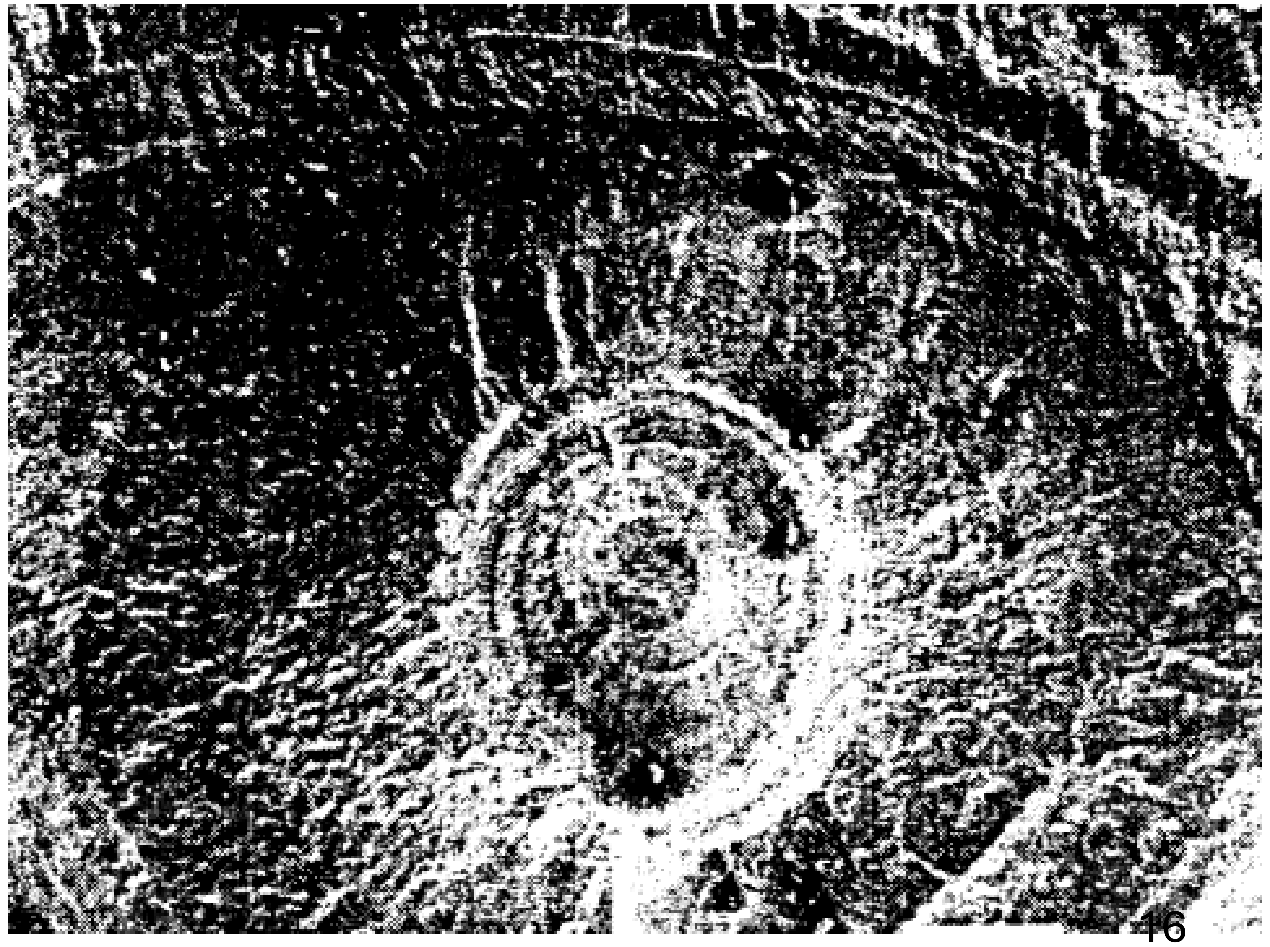


Vertical Split Rims – A portion of the front rim face, or back rim face, fractures off the wheel. Tread damage is often associated with these failures.

WM 66 Cracked or Broken Flange

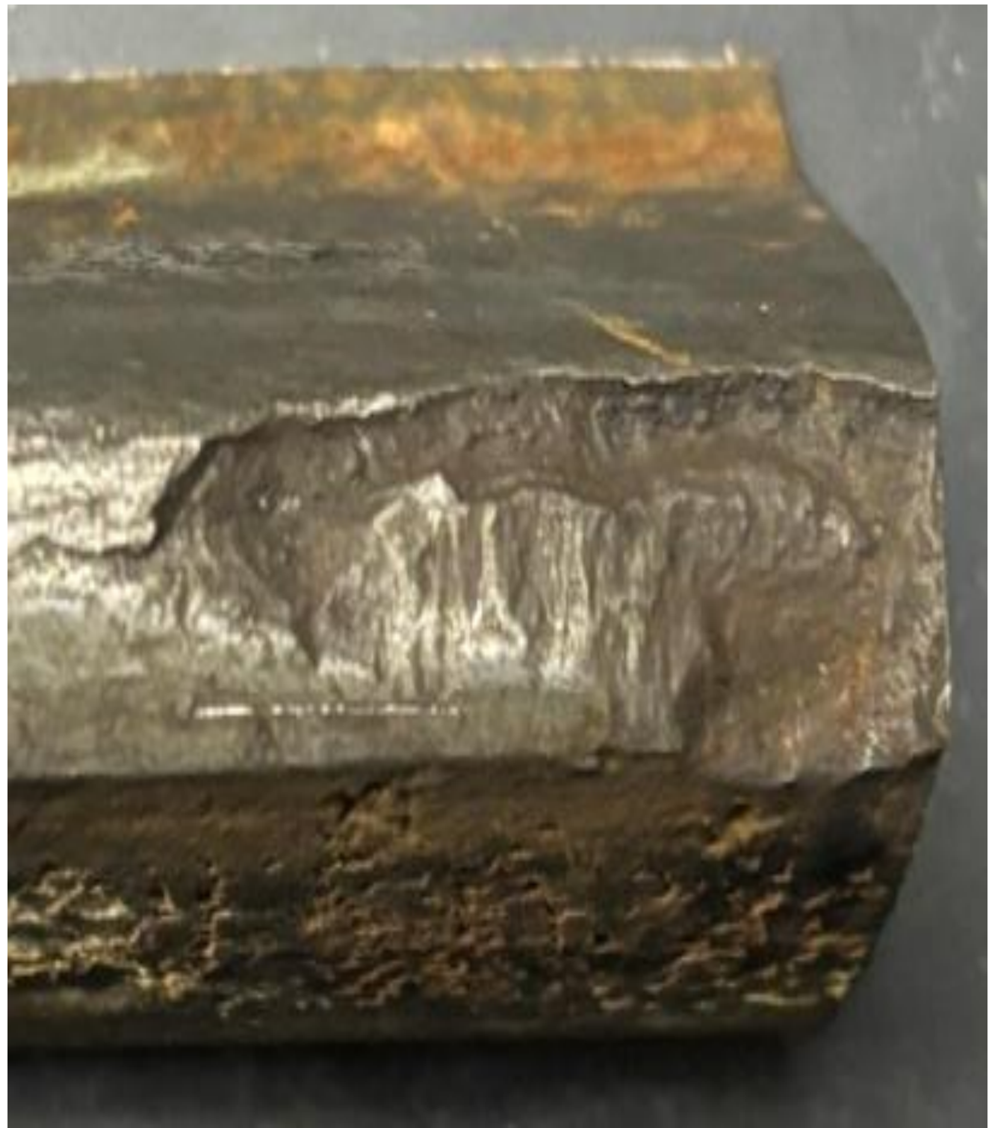
- The broken flange description has - at least two - distinct mechanisms that are now reported within the same WM Code 66
    1. Shelled flange face from RCF – Identified as “Flange Shells” by TTCI at FAST in 1999, along with....
    2. Flange apex-initiated cracking – new NA service phenomenon??
    3. Other?
- Broken flange (WMC 66) is now the most reported code in MD-115
- <u>Recommend creating new Why Made Code(s) to provide better clarity on what is the root cause – similar to what was done with WM 68 / 1D</u>
- Service-related mechanisms have been shown to contribute to the Flange Face Shells – rail lube, wheel/rail profile, etc.

- WM 66 - “Flange Shell” initiated beneath the flange face
- TTCI identified these during 1999 FAST operations
- Main factors:
  - “Dry” FAST operation (rail lubrication halted)
  - High curve service
  - Closed loop
  - RCF / Plastic deformation on the flange face
  - 2-point wheel – rail profile, high force at flange face



Flange Apex Failures

- WM 66 as a failure initiating at the flange apex
- Likely loading at the apex
- Contribution of flange bearing frogs, diamonds?
- Rolled/damaged material at the apex provides a notch for fatigue by both apex and flange face contact fatigue



Various Flange Failures

Flanges Supporting The Freight Car's Load?

- Wheel flanges are supporting the load of the freight car on flange bearing frogs and crossings
- Flange Bearing Trackwork – Began late 2000's, FRA waiver Oct. 2010
- These crossings have led to significant improvements for track maintenance because the wheel tread does not impact the frog, thus leading to fewer frog weld repairs/changeouts, etc.
- However, the flange apex is not designed to support the nominal 35,750 lb. wheel load associated with a 286,000 lb. GRL car
- Also, the new AAR 2A profile is narrower than the old AAR 1BWF
- As the wheel flange gets thinner, the flange/frog stress gets higher with load carried at a sharp tip
- Further investigation needed with flange failures now #1 cause

Flange Bearing Crossing – OWLS – One way low speed

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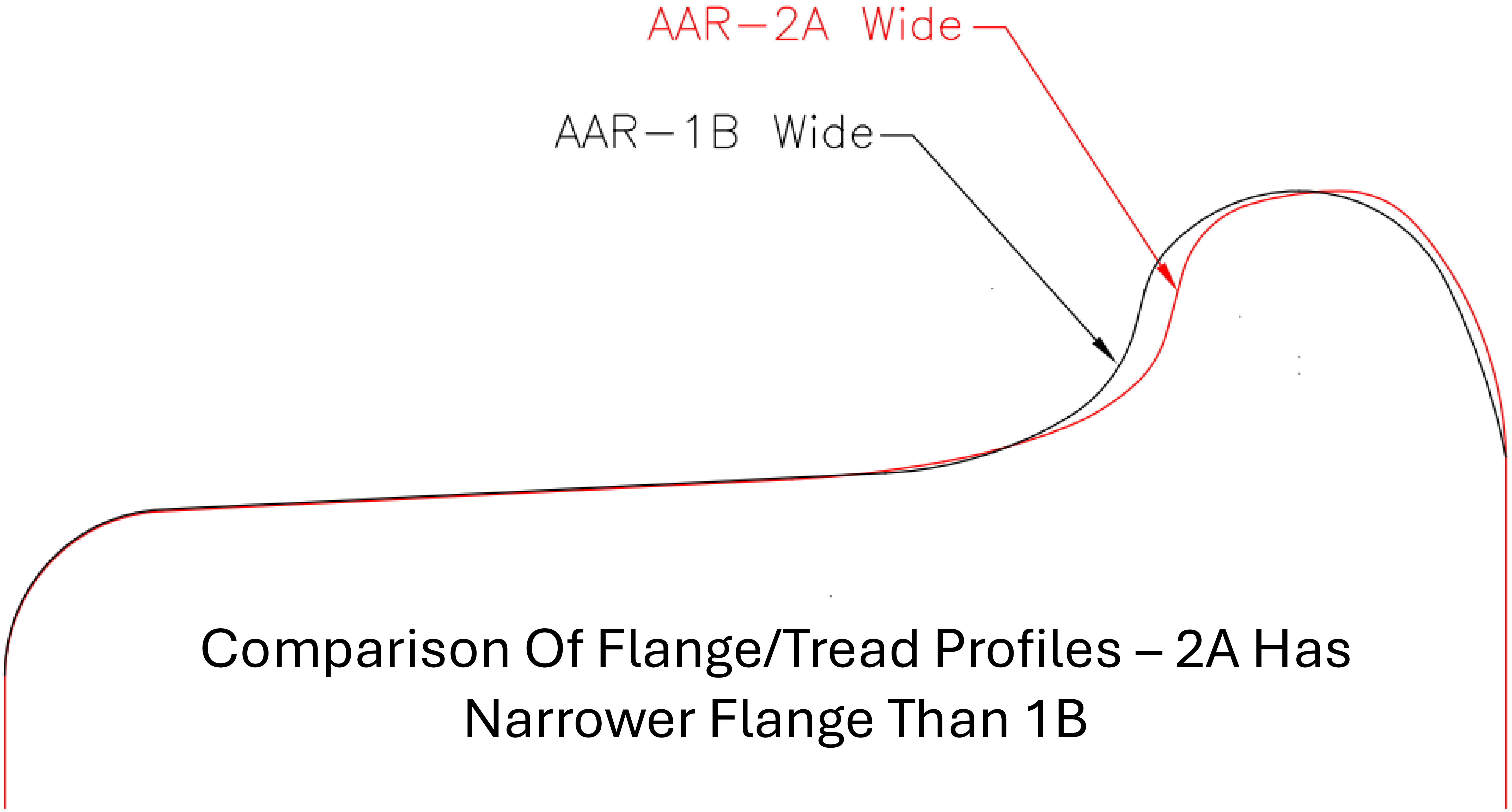
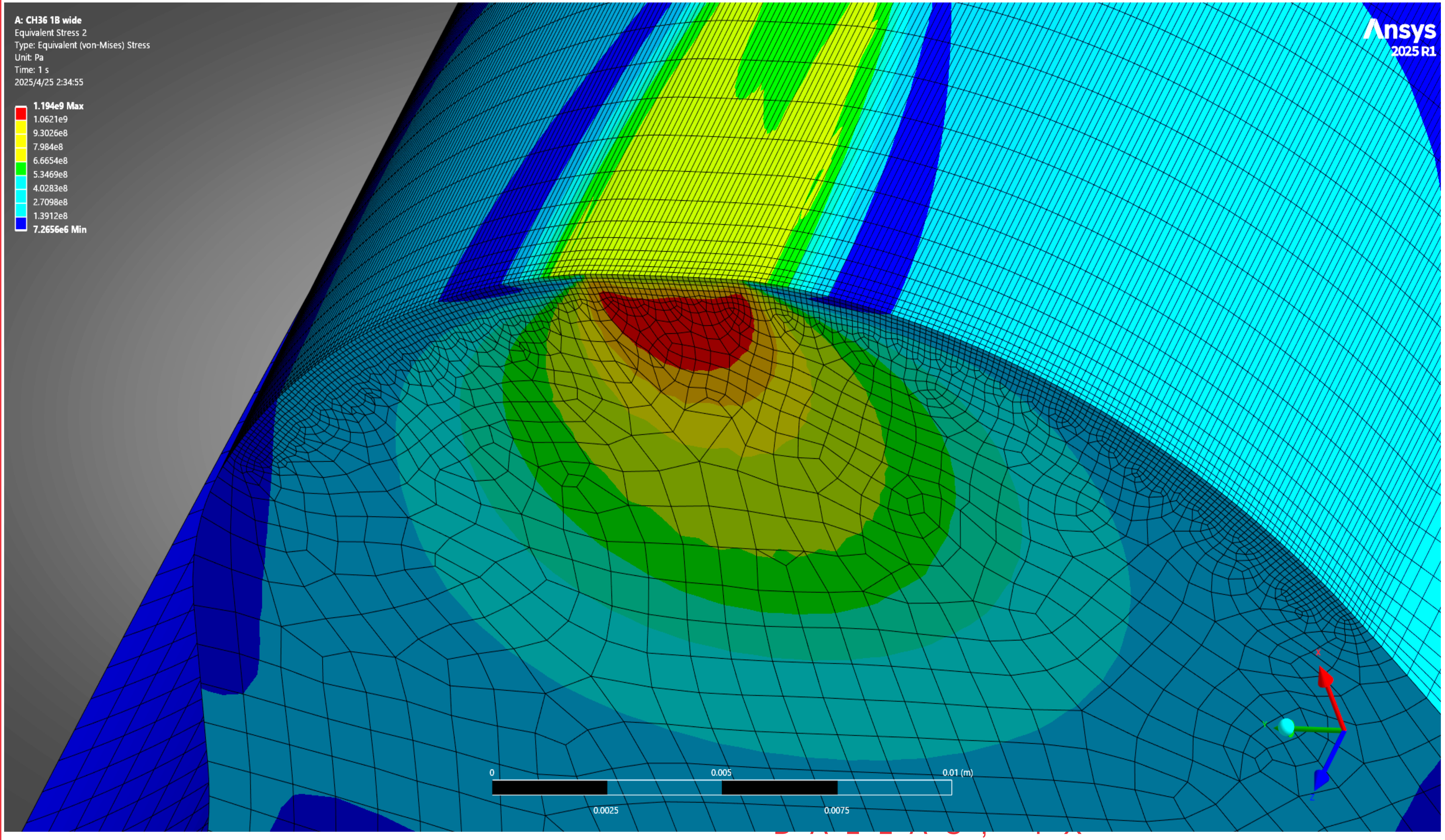


Comparison Of Flange/Tread Profiles – 2A Has Narrower Flange Than 1B

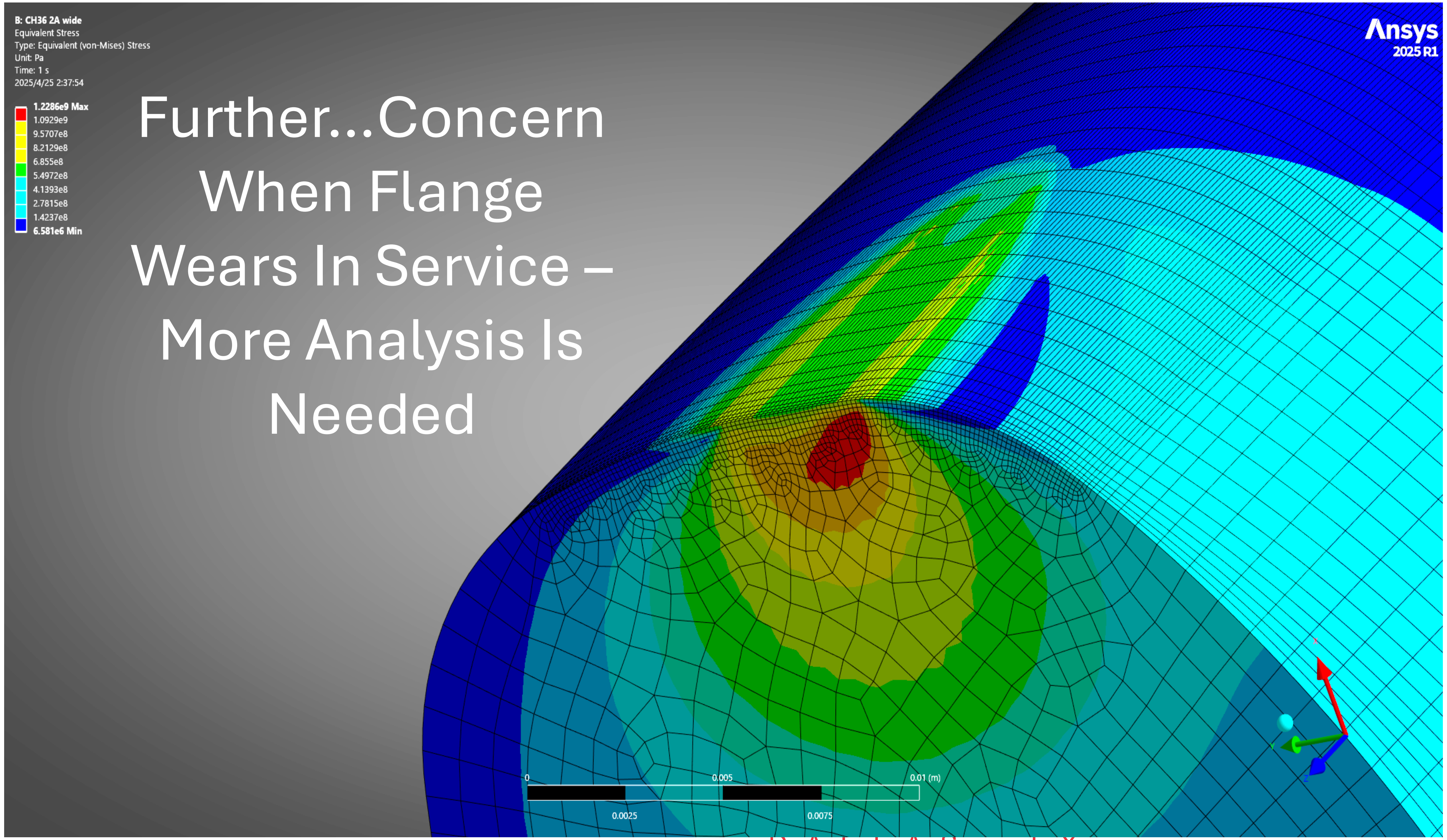
FEA - AAR 1BWF – 35,750 lb. load, flange apex



Max. subsurface shear stress is 685 MPa, or 99.35 ksi

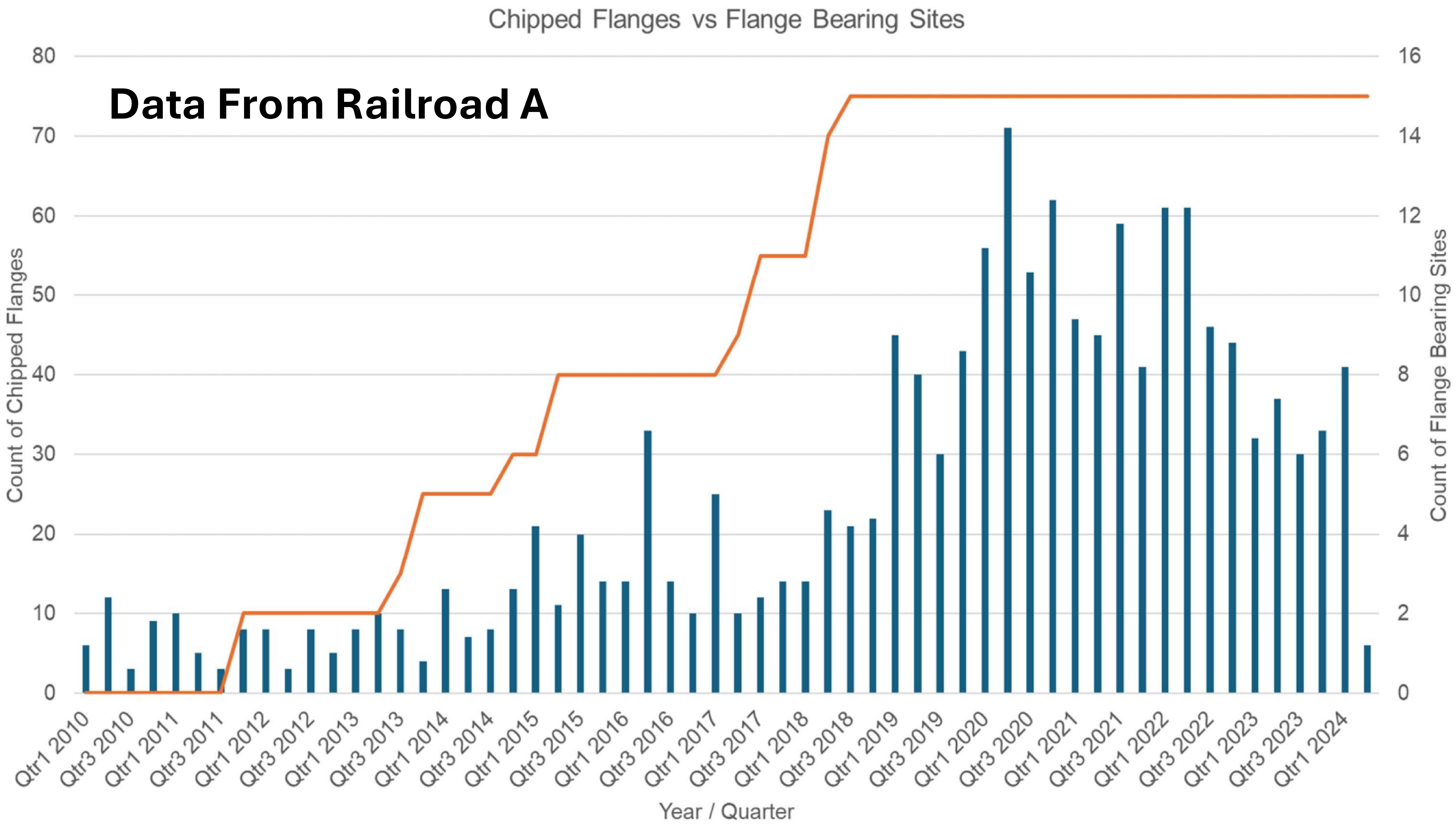
Von Mises stress is 1,194 MPa or 173 ksi

FEA AAR 2AWF – 35,750 lb. load, flange apex



Max. subsurface shear stress is 707 MPa, or 102.5 ksi

Von Mises stress is 1,229 MPa or 178 ksi


Chipped Flanges vs Flange Bearing Sites
Data From Railroad A
Count of Chipped Flanges
Count of Flange Bearing Sites
0
10
20
30
40
50
60
70
80
0
2
4
6
8
10
12
14
16
Qtr1 2010
Qtr3 2010
Qtr1 2011
Qtr3 2011
Qtr1 2012
Qtr3 2012
Qtr1 2013
Qtr3 2013
Qtr1 2014
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Qtr1 2015
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Qtr3 2019
Qtr1 2020
Qtr3 2020
Qtr1 2021
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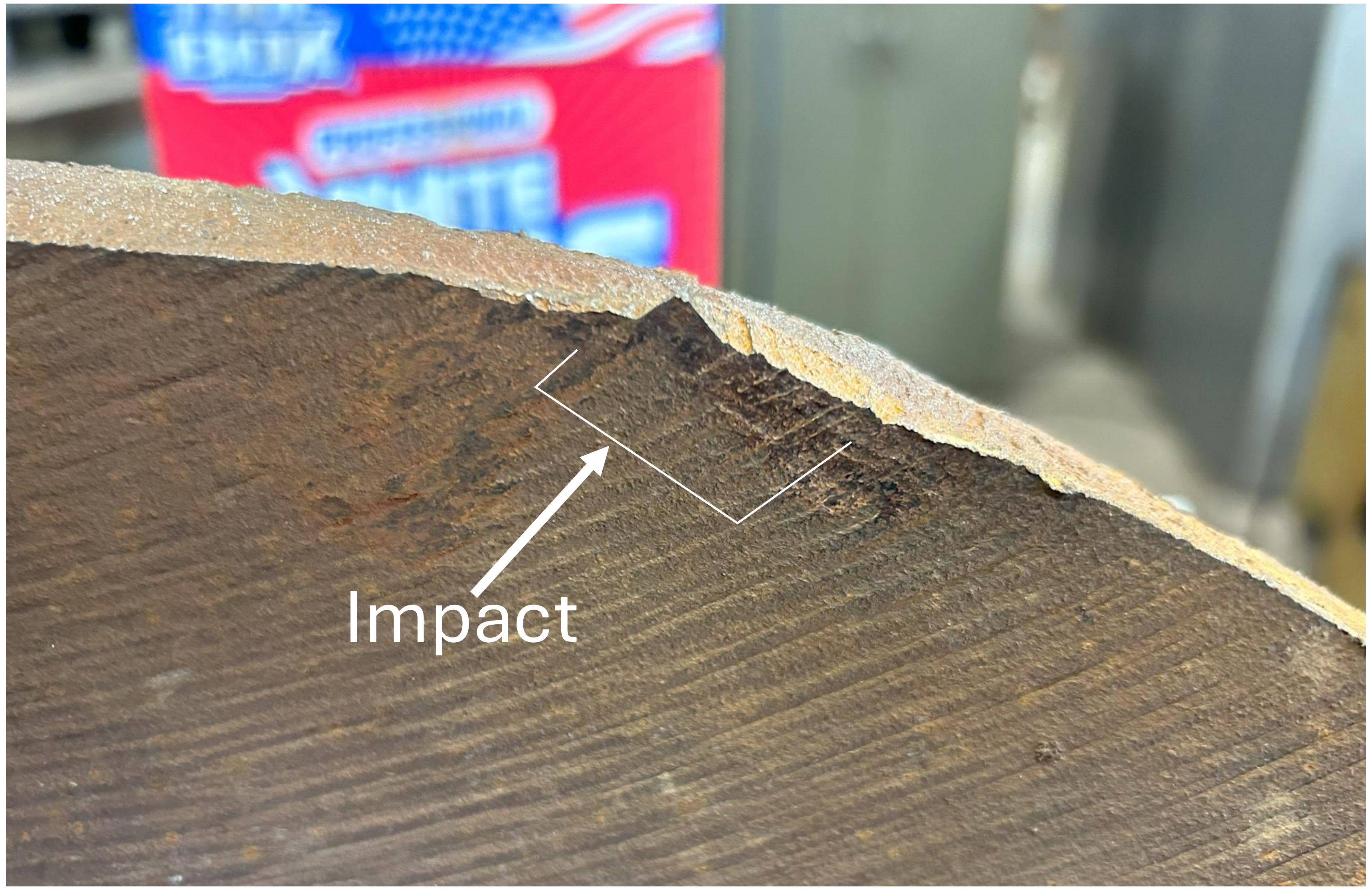
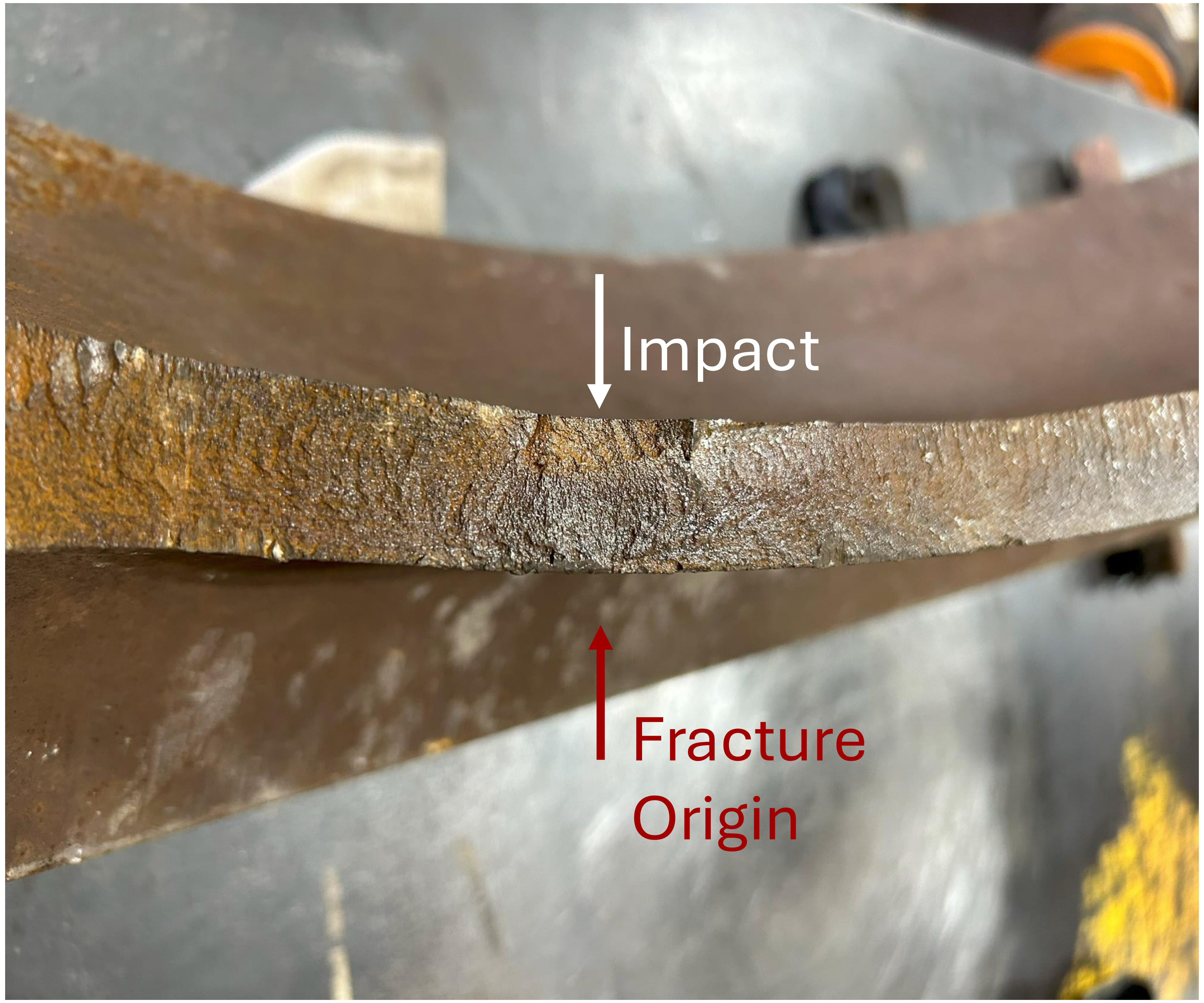
Qtr1 2022
Qtr3 2022
Qtr1 2023
Qtr3 2023
Qtr1 2024
Year / Quarter

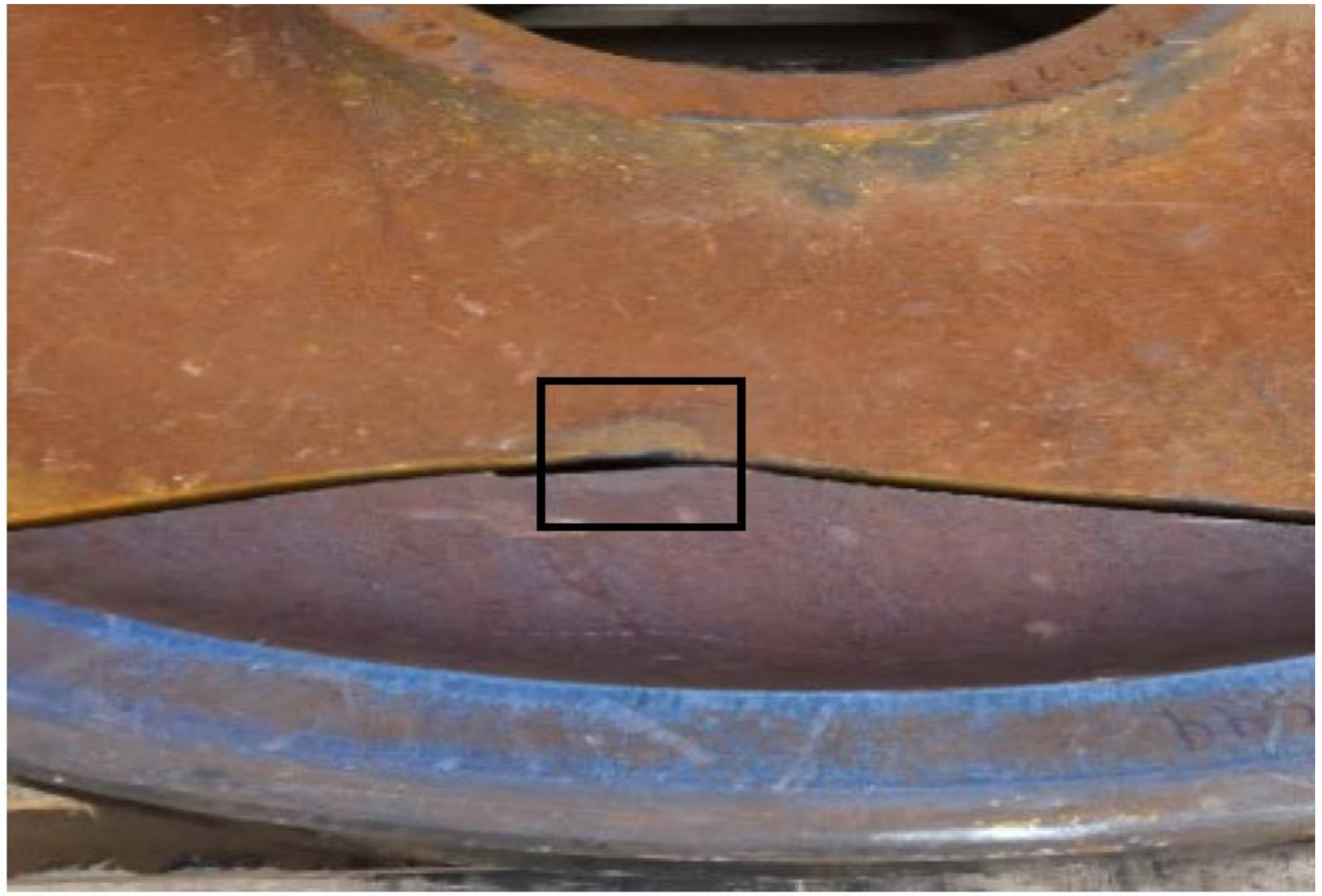
Damage To Wheel Plates Leading to Failure

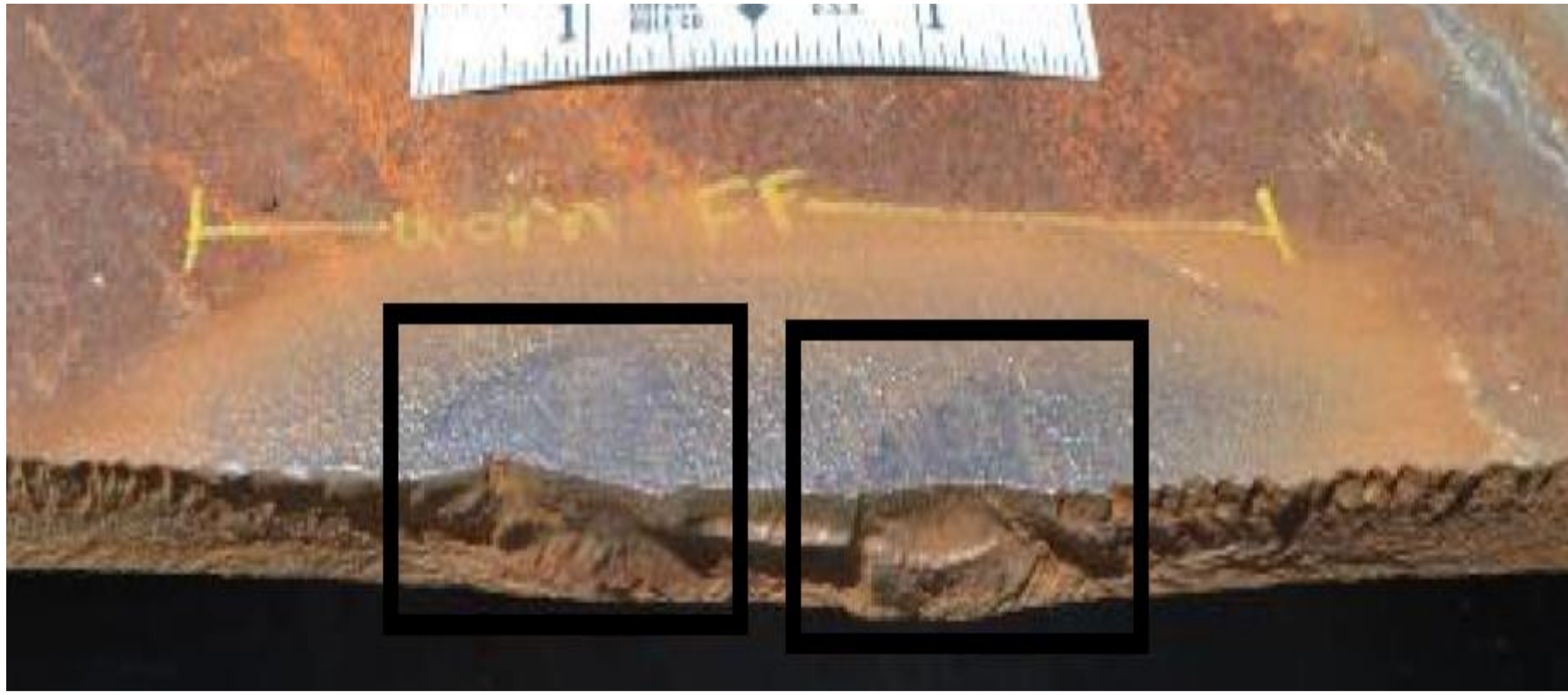


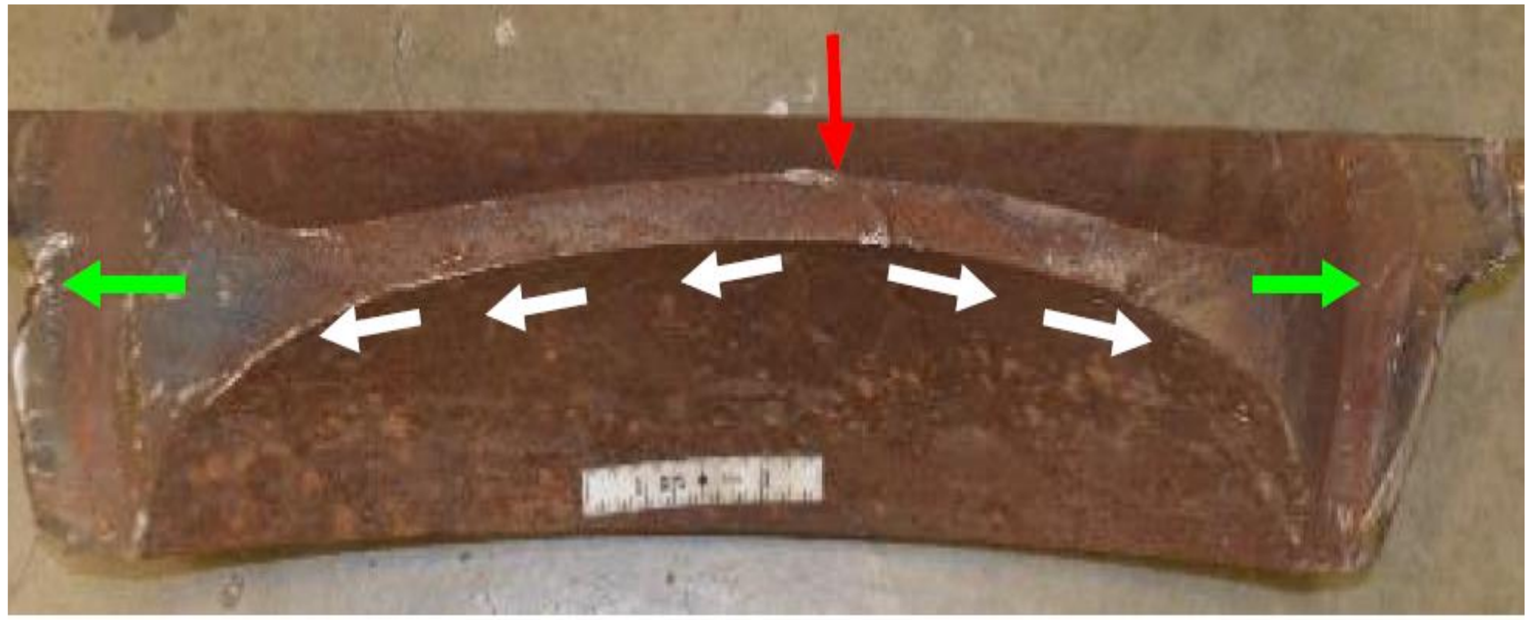
Note the "pressed in" look to the damaged area in above photo. Difficult to Detect. What is causing this?

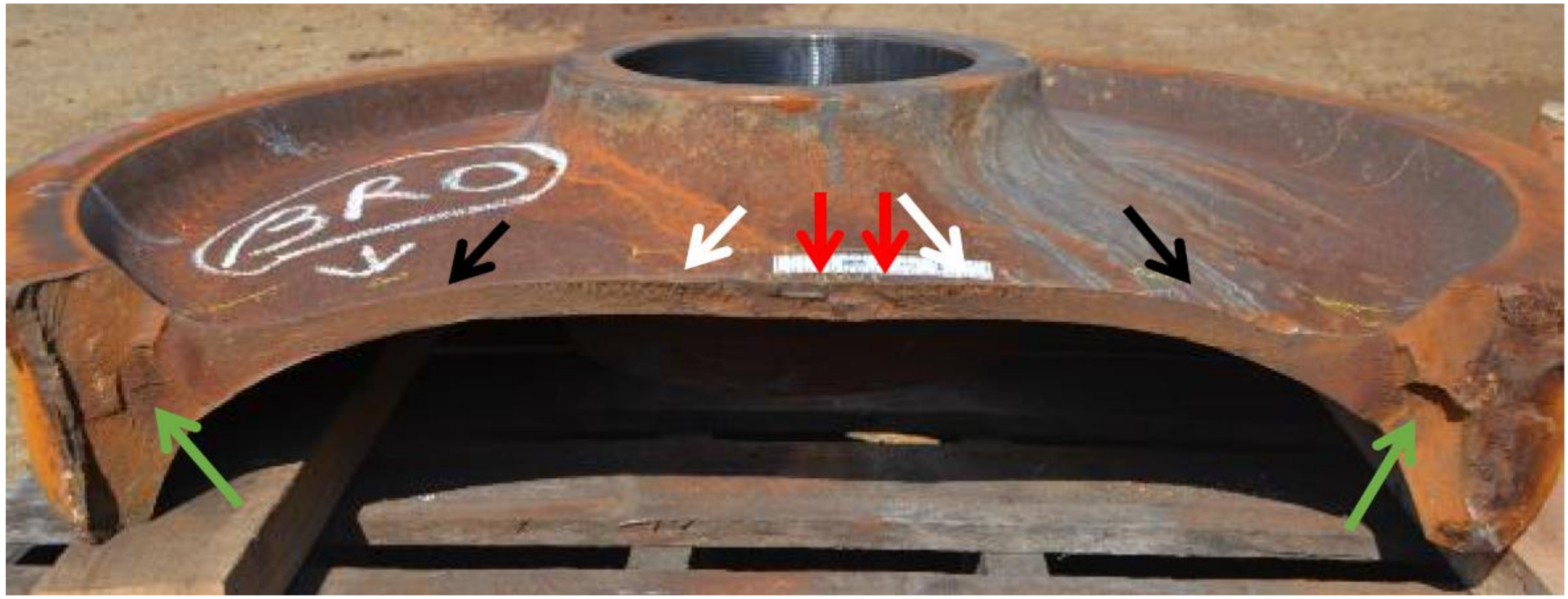


2 Examples Where Plate Damage Led To Failures









Another Example - Plate Failure Due To Impact

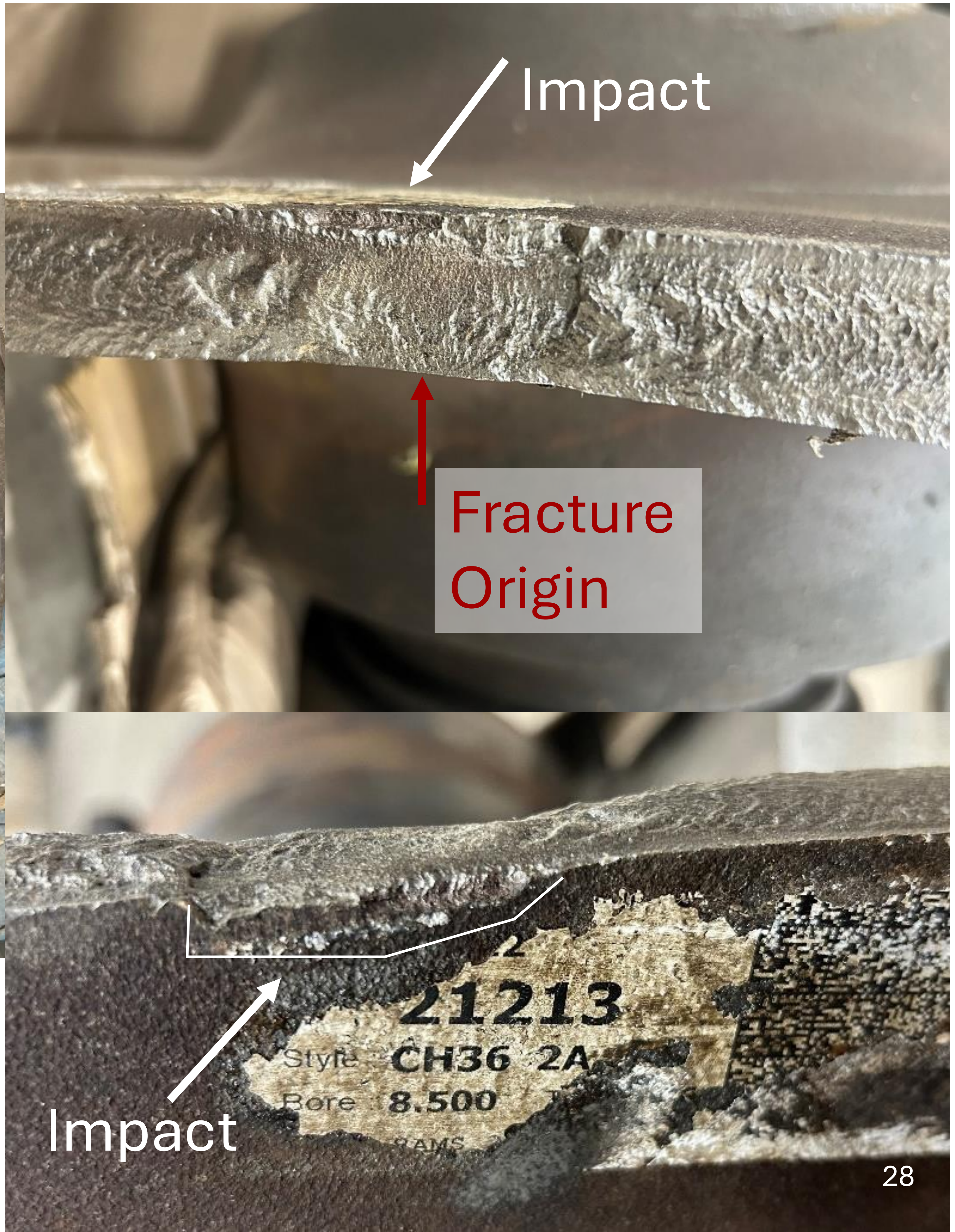


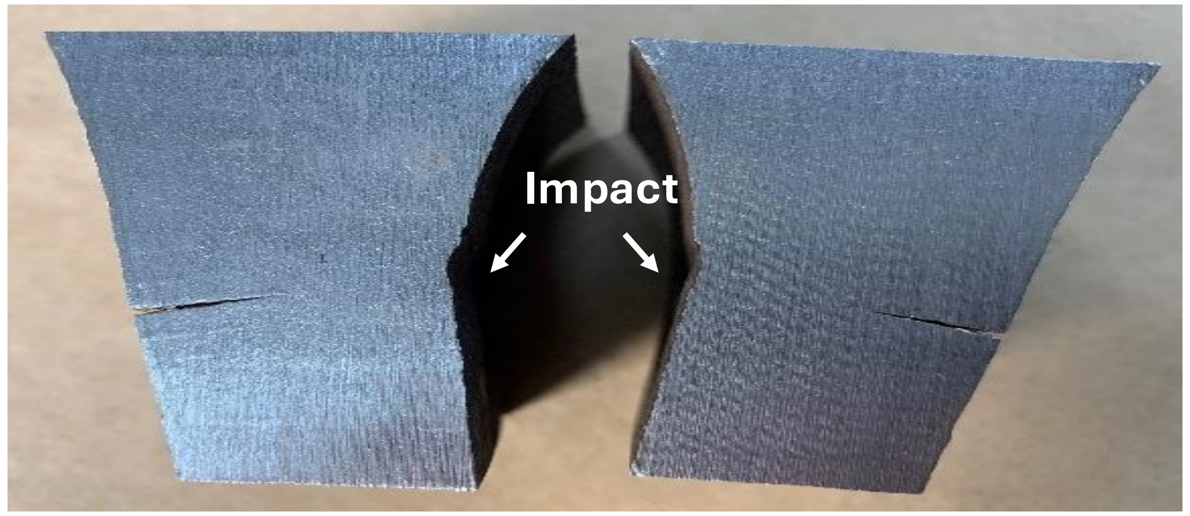
Plate Failures End In Derailment –
Very Dangerous

Further Analysis – Plate Failure

Service- Returned Wheel
GI #22153 Cast 10/03

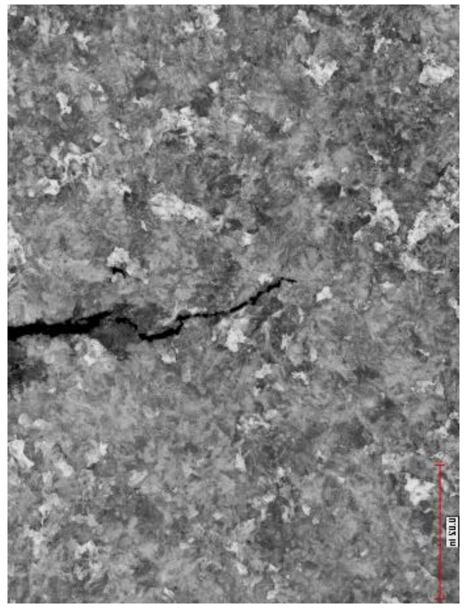
Front Plate Surface

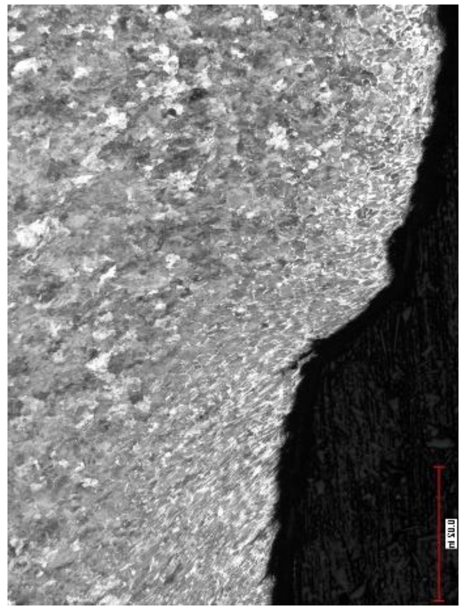
Cross-Section



Back Plate Surface

De-Carburization Absent Along Crack





Etched Surfaces Under Magnification

Wheel Plate Failures - Impact Testing

Impact Testing Plates
at Granite City, IL

Wheel Support Platform Prototype

Drop Hammer

2-in Striker

Compression Load
Tests Also Planned

The industry needs to determine where wheel plate impact damage is occurring.

Shops, repair locations, other?

Locations where “Purchased Core” wheelsets are being harvested? The photo to the left reportedly is such a location.

WABL is planning to look at requirements for purchased cores, to improve safety of Wheels, Axles and Bearings.

Broken Plate Due To Weld Arc Strike

Closer View Of Arc Strike On Plate

Weld Spatter Damage On Wheel Plate

Closer View Of Damage and Plate Cracks

Acknowledgements

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- AAR WABL Committee Members
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- Matt Erickson – Amsted Rail
- Qifeng Guo – Amsted Rail